arXiv:2609.34309v1 [cs.CV] 28 Sep 2026

MaLiang-Harness: A Programmable Path to Image and Video Generation

Haoyu Zhao[1,2,†], **Zihao Zhang**[1,2], **Xudong Wang**[1], **Jiaxi Gu**[3], **Zuxuan Wu**[2], **Yu-Gang Jiang**[2], **Shuicheng Yan**[1]

[1]National University of Singapore [2]Fudan University [3]Tencent

Abstract

Executable programs offer explicit control over how images and videos are constructed, but generating runnable code is only the beginning of visual creation. A program can execute correctly while violating the requested composition, appearance, or motion. We define this discrepancy as the Program-to-Visual (P2V) gap and introduce MaLiang-Harness, a unified framework for organizing MLLM-driven visual generation into a persistent process of construction, inspection, and revision. Its central design is to make the evolving visual program, its construction history, and its verification share a common revision reference. We define the Persistent Executable Generation (PEG) state as preserving programs and task context. Traceable Generation Process (TGP) connects edits to rendered evidence, and Revision-aware Editing and Verification (REV) supports restoration and checks the current revision before completion. Together, these mechanisms coordinate planning, execution, and visual feedback across rendering backends. We evaluate 11 powerful closed-source MLLMs on MaLiang-IBench and four on MaLiang-VBench, measuring generation success, visual quality, and computational cost. GPT-6-Astra achieves 100% generation success on both benchmarks, with 96.0% of image tasks and 76.9% of video tasks meeting all quality thresholds. The comparison also reveals a mismatch between general capability scores and visual generation performance, with similarly scored models differing substantially in their ability to satisfy visual requirements. MaLiang-Harness provides a systematic basis for studying how MLLMs translate executable code into visual outcomes, exposing both the potential of programmable generation and the limitations of general benchmarks as predictors of this ability. The project is available at https://github.com/gulucaptain/MaLiang-Harness.

1 Introduction

"The idea becomes a machine that makes the art."

— Sol LeWitt

Over the past decade, advances in high-quality image and video generation have largely followed the paradigm of direct visual synthesis. From generative adversarial networks to diffusion- and flow-based models [Ho et al., 2020, Lipman et al., 2022], these approaches learn the distribution of visual data and directly synthesize pixels or latent visual representations [Rombach et al., 2022]. Although this paradigm has achieved remarkable success, its underlying generation process typically remains implicit. In contrast, the rapidly improving capabilities of multimodal large language models (MLLMs) in semantic understanding, reasoning, and code generation are making a programmable path to image and video generation increasingly viable: **a model expresses creative intent as an executable visual program, which a renderer subsequently converts into an image or video.**

[†]Project lead.

Table 1: **Comparison of creative-style coverage and generation capabilities.** We qualitatively compare MaLiang-Harness with existing image (T2I) and video (T2V) generation models along two dimensions: 1) creative-style coverage spanning line and flat art, 2D cartoons, pixel art, stylized 3D, mixed media and craft, and photorealism; and 2) generation capabilities including controllability over visual content and spatiotemporal structure, editability of the generated artwork, and traceability of the creation process. MaLiang-Harness supports diverse visual styles through executable visual programs, while providing explicit control, targeted editing, and an inspectable construction process. ● Supported. ◐ Partially supported. ○ Not established.

Type	Method	Line&Flat Art	2D Cartoon	Pixel Art	Stylized 3D	Mixed & Craft	Photo-realistic	Contro-llability	Edita-bility	Process Traceability
Image	T2I	●	●	●	●	●	●	●	●	○
	Ours	●	●	●	●	●	◐	●	●	●
Video	T2V	●	●	●	●	●	●	●	●	○
	Ours	●	●	●	●	●	◐	●	●	●

However, a program can execute correctly yet produce an image or video that fails to satisfy the user's intent. An object may appear in the wrong place, or an animated event may occur at the wrong time. We refer to this discrepancy between program-level correctness and visual requirement satisfaction as the Program-to-Visual (P2V) Gap. Bridging this gap requires the model to reason about the visual consequences of its code. Its initial plan must account for the choice of visual representation and rendering backend. Once the code is executed, the rendered output provides evidence for revising both the program and the plan. This process must preserve the artwork across iterations so that the model can address unmet requirements while retaining access to earlier versions. Each revision also calls for renewed visual assessment because a change can affect requirements that were previously satisfied. These demands motivate a stateful harness that connects creative intent to executable visual representations and supports continued control over the artwork through visual feedback.

To address this challenge, we introduce **MaLiang-Harness**, a unified interface connecting creative intent to multiple executable visual representations. It unifies programmable image and video generation through a shared protocol for state management, rendering, and revision-aware verification. Through this interface, an MLLM selects suitable rendering backends and exercises explicit control over the artwork through visual feedback. Code directly specifies how visual content is constructed, from spatial composition to motion over time. Optional image assets complement this programmatic structure when rich appearance is difficult to express through code alone. Images and videos share the same creation framework: at a fixed program revision, an image is rendered at a specified content time, while a video samples the temporal behavior encoded by the program. Table 1 summarizes the creative-style coverage and functional capabilities.

Furthermore, MaLiang-Harness organizes creation into a loop in which the MLLM plans and generates code, then uses execution results and visual feedback to guide revision. Three designs support this process: **1)** *Persistent Executable Generation (PEG) State* maintains the executable definition of the evolving artwork across iterations. It brings programs and assets together with their spatiotemporal organization, task requirements, the current plan, and revision index. This gives the model a stable object to inspect and modify as creation progresses. **2)** *Traceable Generation Process (TGP)* makes the construction of the artwork inspectable through explicit programs and recorded operations. Intermediate renderings expose the visual effects of these operations, helping the model and the user relate an observed discrepancy to the components or changes that may have produced it. **3)** *Revision-aware Editing and Verification (REV)* allows the model to restore earlier artwork content as a new revision while retaining current task requirements. Observations and verification results are associated with specific revisions, and every committed update requires renewed assessment before completion. We demonstrate that these designs enable the MLLM to translate visual feedback into targeted revisions of the evolving artwork, helping bridge the P2V gap.

We evaluate 11 MLLMs on MaLiang-IBench and four on MaLiang-VBench. GPT-6-Astra [OpenAI, 2026b] achieves a 100% generation success rate on both benchmarks. Its outputs meet every quality threshold on 96.0% of image tasks and 76.9% of video tasks, compared with 86.0% and 38.5% for